D4 Data errors

Row counts across layers, distribution drift and weight totals.

D4-L1 Row counts across layers

Write the expected relationship between layers as rules, compute counts in one pass on identical filters, and halt when a CRITICAL rule fails.

Row-count rules: silver vs bronze	Silver rows = distinct respondent keys in bronze for the tracker and wave.
Rows after a MERGE	Rows before + numTargetRowsInserted (minus any deletes). Updates add nothing.
Counting efficiently	One agg pass: F.count('*') and F.count_distinct(key). Not repeated count() calls.
The get_unprocessed_waves bug	Bronze filtered on tracker, silver not. Filter both sides on the same keys.

D4-L2 Distribution drift between waves

Compare shares over the union of categories, flag new and missing values, summarise with PSI, and guard every total.

```
PSI = sum over categories of (actual - expected) x ln(actual / expected)
```

Three drift checks	Category shares vs prior wave, new and missing categories, and a PSI summary.
PSI formula	Sum of (actual - expected) x ln(actual / expected) over shares, with zero shares floored.
PSI thresholds 0.1 and 0.25	A rule of thumb, not a standard. Calibrate on your own wave history.
Percentage points vs relative change	30% to 45% is +15 points and +50% relative. Say which a tolerance uses.

D4-L3 Weight totals and survey checks

Check nulls, range, sum and efficiency per wave and subgroup. Mind Spark's nulls: sample stddev, empty aggregates and sums over no matches.

```
deff  = n x sum(w^2) / (sum w)^2 = 1 + CV^2
n_eff = (sum w)^2 / sum(w^2) = n / deff
```

Kish design effect and effective sample	deff = n x sum(w^2) / (sum w)^2 = 1 + CV^2. n_eff = (sum w)^2 / sum(w^2).
F.stddev in Spark	Sample standard deviation, null for one row. Use F.stddev_pop for 1 + CV^2.
sum(when(cond, 1)) with no matches	Null, not 0. Wrap in coalesce(..., lit(0)) or add otherwise(0).
Severity for a null weight	CRITICAL: halt before gold.